Visual Deformation for Pseudo-Haptic Softness Perception in Optical See-Through Augmented Reality

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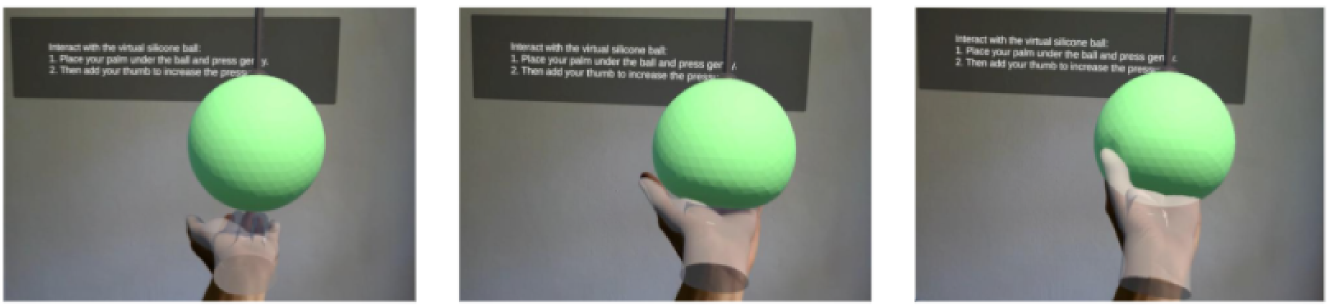

Fig. 1: The presented experiment evaluated the sense of softness during interaction with a soft ball in an AR environment. The participants performed a series of actions: put the palm under the ball and add the thumb.

***Abstract*—Can visual manipulation alone induce a convincing pseudo-haptic perception of softness during bare-handed interaction in mid-air AR? This pilot study investigates human perception without the use of physical objects or active tactile actuators. In an augmented reality environment, users interact with a soft virtual ball using a natural semi-squeeze gesture based on the palm and thumb. By manipulating the Control/Display (C/D) ratio of the visual deformation of the object, we modify the visual feedback associated with the user's physical hand movement. Our results suggest that visual manipulation alone can significantly alter perceived softness. Furthermore, the study highlights current hardware limitations, showing how occlusions and instability in hand tracking can break visuo-motor coherence and disrupt the pseudo-haptic illusion.**

***Index Terms*—Pseudo-haptic feedback, Augmented Reality, Visuo-proprioceptive conflict, Human-Computer Interaction, Softness perception.**

I. Introduction

Augmented Reality (AR) offers the possibility of overlaying digital content onto the physical world, enabling new forms of natural interaction. Modern head-mounted displays (HMDs) increasingly allow for bare-hand mid-air interaction, reducing the need for physical controllers while still maintaining a good level of tracking accuracy [1].

This pilot study investigates whether visual manipulation alone can induce a **pseudo-haptic perception of softness** during interaction with a purely virtual object, and therefore with an object that is physically intangible. To achieve this goal, we rely on the principles of pseudo-haptic feedback [2]. According to Bayesian Multimodal Integration models (BMI) [3], [4], the human brain combines sensory signals by weighting them according to their reliability. In spatial interactions, vision often has a strong influence over proprioception. By exploiting this **visual dominance**, controlled **visuo-proprioceptive conflicts** can be created to alter the perceived haptic properties of the user.

Building on the studies of Kawabe [5], which demonstrated the feasibility of stiffness illusions in mid-air actions, this research extends the concept to direct spatial interaction in AR environments. However, the use of an Optical See-Through (OST) headset, such as the Magic Leap 2 used in this study, introduces a fundamental constraint. The interaction happens in mid-air and the virtual ball does not provide a real physical boundary. Therefore, the user can move the hand freely through the virtual object, because there is no physical surface that can stop or constrain the movement.

This constraint makes direct manipulation of the hand difficult. Unlike previous pseudo-haptic experiments based on a projected or virtual hand, it is not possible to visually block, slow down, or shift the hand movement in a strong way without breaking the coherence of the interaction [6]. The Magic Leap 2 dimming feature was used to reduce the visibility of the real hand and the surrounding environment as much as possible, but it does not create a physical constraint.

For this reason, our experiment shifts the focus of manipulation exclusively onto the object, a virtual soft ball. By altering the Control/Display (C/D) ratio between the actual physical semi-squeeze movement, based on palm and thumb interaction, and the extent of the visual deformation of the virtual ball across three gain levels (low, medium, and high), the study aims to evaluate whether visual deformation alone can generate a convincing pseudo-haptic illusion of softness.

II. Related Work

In this section, we review works related to ours, analyzing the approaches that are useful for answering our research question.

A. Pseudo-Haptic Feedback and Visual Dominance

Pseudo-haptic feedback is based on the idea that visual feedback can influence the perception of haptic properties, even when no physical force feedback is provided. As shown in previous studies, when sensory information is ambiguous or conflicting, vision can have a stronger influence on perception [7]. This is important for our study because the illusion depends on the relation between the real movement of the user and the visual response of the virtual object.

However, users do not respond uniformly to pseudo-haptic feedback. Lécuyer et al. show that there can be different user profiles, such as visually oriented users, haptically oriented users, and users who are sensitive to both visual and haptic information [8]. Their work also uses psychophysical measures such as the Just Noticeable Difference (JND), the Weber Fraction, and the Point of Subjective Equality (PSE) to study the boundary of the illusion. In our pilot study, JND and PSE are not directly estimated, because the experiment uses fixed deformation gain levels and subjective ratings. However, these measures are useful to understand how pseudo-haptic perception can be studied in future versions of the experiment.

B. Mid-Air Interaction and Compliance Perception

A useful concept for our study is deliberate *user priming*. Pusch and Lécuyer describe user priming as one of the elements that can influence pseudo-haptic perception, because the expectations of the user can affect how visual and haptic information are interpreted [4]. This relates to the well-established idea that a **visual appearance** of the object shapes expectations about its **physical properties**. For example, Weiss, Villa, Ziarko, and Muller show that stiffness perception can be influenced by the visual behavior of compliant objects during interaction [9]. In our experiment, the virtual object is presented as a soft ball. This visual context, together with the information given to the user about the presence of multiple trials with different levels of elasticity, can guide the user to interpret the deformation of the object as softness or compliance. For this reason, user priming is relevant, but it is not used as the main manipulation. The main manipulation remains the visual deformation gain applied to the ball.

Another important concept is the manipulation of the Control/Display (C/D) ratio. The C/D ratio describes the relation between the physical movement of the user and the visual response shown by the system. Samad, Gatti, Hermes, Benko, and Parise showed that the perceived weight of virtual objects can be changed by modifying the C/D ratio [10]. Their study focuses on weight perception, but the same principle is useful for our work because our experiment modifies the relation between the semi-squeeze movement and the visual deformation of the ball. This pilot study therefore focuses on creating a visual displacement and deformation in the virtual ball by modifying the C/D ratio. The goal is not to generate a real physical force, but to create a visual response that can suggest softness without breaking the interaction of the user.

C. Constraints of Optical See-Through AR

Optical See-Through AR introduces specific constraints for pseudo-haptic feedback. In VR or projected-hand experiments, the system can often control the visual representation of the hand and create stronger visual conflicts. In our case, however, the interaction happens in mid-air and there is no physical object that constrains the hand movement; the user can move freely through the virtual object, because the ball has no real physical boundary.

For this reason, directly manipulating the hand movement would be difficult. The system cannot physically stop the hand, and a strong visual manipulation of the hand could break the coherence of the interaction. The Magic Leap 2 dimming feature is used to reduce the visibility of the real hand and the surrounding environment as much as possible, but it does not create a physical constraint. Therefore, the main limitation is not only the visibility of the real hand, but also the absence of real spatial and physical constraints during mid-air interaction.

The work of Argelaguet, Hoyet, Trico, and Lécuyer is useful to understand the role of virtual hand representation in agency and ownership [11]. Their study shows that the way the hand is represented can influence the experience of the user. In our case, however, the prototype does not rely on a fully virtual hand. The hand is tracked by the headset, while the visual manipulation is applied mainly to the virtual object. This choice helps preserve the coherence between the real movement of the user and the visual response of the ball.

Temporal coherence is also important. Knorlein, Di Luca, and Harders showed that visual and haptic delays can influence stiffness perception in augmented reality [12]. Their study focuses on stiffness, but it is relevant for our work because it shows that the timing between movement and visual feedback can change the perceived material properties of an object. In our prototype, the deformation of the ball must therefore remain synchronized with the hand movement to preserve the pseudo-haptic illusion.

III. Experimental Methodology

The modeling of the experiment moved across three main areas: the sense of agency during the interaction, the visual mismatch created through the deformation of the ball, and the role of the Control/Display (C/D) ratio.

A. Sense of Agency and Interaction Coherence

The Magic Leap 2 includes an internal hand tracking system, so it was not necessary to add a virtual mesh or create a fully virtual hand model. The interaction is based on the real movement of the hand, detected directly by the headset.

The real hand moves freely in space, while the Magic Leap 2 tracks the palm, fingers, and thumb in real time. When the tracked hand reaches the virtual ball, the ball reacts in real time to the movement. In this way, the user can see a direct relation between the physical gesture and the visual response of the object.

The contact proxies inserted on the fingertips help the interaction appear less free and less disconnected. They support the relation between the hand movement and the deformation of the ball, so the visual response follows the gesture of the user. However, in this study, it is more correct to talk about a sense of agency rather than ownership in the strict sense. The user does not control a fully virtual hand, but the system creates coherence between the real hand movement and the visual behavior of the virtual object.

B. Visual Mismatch

The visual mismatch is created by modifying the deformation of the virtual ball. Through the headset tracking, the Unity system reads the hand gesture and calculates normalized values related to semi-squeeze intensity and thumb contact.

These values are then used to estimate the contact or pressure of the hand on the ball. A deformation script modifies the mesh of the ball, creating compression in the contact area and lateral bulging on the surrounding surface. The deformation is updated in real time, following the gesture of the user.

The experimental levels low, medium, and high are applied by increasing or decreasing how much the ball deforms for the same hand movement. Therefore, the deformation is **not physically simulated**, but visually controlled to produce a consistent pseudo-haptic response. It is a controlled visual deformation, guided by hand tracking data and used to create a **pseudo-haptic impression of softness**.

C. Role of the C/D Ratio

The Control/Display (C/D) ratio is used to define the relation between the real movement of the hand and the visual response of the ball. In this experiment, the control component is the actual movement of the hand, mainly the palm and thumb movement. The display component is the amount of visual deformation shown on the virtual ball.

When the same hand movement produces a small deformation, the visual gain is low. When the same movement produces a stronger deformation, the visual gain is high. In this way, the system can create different levels of perceived softness without changing the real physical movement of the user.

The role of the C/D ratio is therefore central to the experiment. It allows the same semi-squeeze gesture to produce different visual responses, making it possible to test whether visual deformation alone can influence the perception of softness or compliance.

IV. Experiment Design

The experiment consisted of six trials, two for each level of deformation gain: low, medium, and high. The trials were preceded by a calibration phase and followed by a post-experiment questionnaire.

The test was conducted on 13 participants aged between 22 and 28 years. Among them, 2 participants were left-handed, while 11 participants were right-handed. Most participants had little or no previous experience with Augmented Reality.

Unity 2022 was used to create the development environment. The headset used for the experiment was the Magic Leap 2, with native OpenXR hand tracking. To focus the attention of the participant on the hand-object interaction, the environment was dimmed to the maximum. This reduced the visibility of the real hand and the surrounding environment as much as possible.

Before the experiment, each participant completed a calibration phase. During this phase, the system collected useful hand tracking data, such as the distance between the thumb and the palm and the squeeze movement. These values were collected through the repetition of simple gestures. In the same phase, the participant answered two personal questions: *"What is your dominant hand?"* and *"How much experience do you have with Augmented Reality?"*. These questions were used only to describe the sample. Being a pilot study, only one hand was implemented: the left; regardless of the user's dominance.

After the calibration phase, the trials started. The deformation gain levels were presented in the following fixed sequence: medium, low, high, low, high, and medium. Considering

$$C/D = \frac{\text{DeformationShown}}{\text{PhysicalCompression}}$$

where $\text{DeformationShown} = \text{TrackingCompression} \times \text{Gain}$, with $\text{Gain} \in \{0.6 \text{ (low)},\ 1.0 \text{ (medium)},\ 1.4 \text{ (high)}\}$.

Each participant had 13 seconds for each trial to interact with the virtual soft ball and to perceive its deformation. At the end of each trial, the participant was asked to answer the question: *"How clearly did you perceive the deformation of the ball?"*. The answer was given using a 7-point Likert scale, where 1 meant "not at all" and 7 meant "very clearly".

At the end of the six trials, the participant completed a post-experiment questionnaire. The questions were:

- Q1 – How soft did the virtual ball feel?
- Q2 – How elastic did the virtual ball feel?
- Q3 – How believable was the contact with the ball?
- Q4 – Did adding the thumb make the interaction feel more convincing?
- Q5 – Did the visual deformation influence your perception of touch?
- Q6 – Did the interaction feel more like touching an object than watching an animation?

- Q7 – How well did the visual deformation match your hand movement?
- Q8 – At any point, did the visual deformation feel disconnected from your hand movement?

Questions Q1 to Q7 used a 7-point Likert scale. Instead, question Q8 was introduced to verify the threshold for the limit of the illusion and used a Yes or No answer. If the participant answered Yes, a follow-up question asked when the disconnection occurred. The possible answers were: *"From the beginning"*, *"First contact"*, *"Adding the thumb"*, and *"Toward the end"*.

Inspired by the staircase procedure toolkit proposed by Zenner et al. [13], a custom Unity toolkit was created to manage the experiment and export the responses directly in CSV format. In this way, it was easier to collect, visualize, and analyze the data obtained from the experiment. However, the present pilot study did not use an adaptive staircase procedure. The deformation gain levels were presented in a fixed sequence.

V. Results

The results extracted from the experiment were divided into three categories:

- pre-data: data related to calibration and participant information, such as dominant hand and previous experience with Augmented Reality;
- trial-data: data related to each trial, linked to the interaction with the virtual ball;
- post-data: data extracted from the questionnaire completed after the soft ball trials.

The pre-data were used to describe the sample. The trial-data were used to evaluate how the visual deformation gain, interpreted as a manipulation of the C/D ratio, influenced the perceived deformation of the ball. The post-data were used to obtain a more general evaluation of the whole experiment and to understand whether the illusion was perceived as convincing.

By visualizing the mean ratings of the trial-data, it is possible to observe that the high deformation gain produced the clearest perceived deformation according to participant ratings.

In the graph, some trials show results far from the average. This demonstrates the variability of the sample. Moreover, in some cases, the same participant evaluated the same deformation level differently across repetitions. This can be interpreted in different ways. First, the participant did not know which condition was presented and could perceive each trial as a new interaction. Second, the arm remained raised during the experiment, so fatigue could have influenced the rating. Finally, since most participants were right-handed, using the less dominant hand could have influenced the perception of the interaction.

To evaluate whether the deformation gain had a significant effect on perceived deformation clarity, a Friedman test was used. This test was selected because each participant experienced all three conditions: low, medium, and high. The Friedman test does not work directly on the original values, but on the ranks assigned to the three conditions for each participant. It checks whether one condition tends to receive higher ranks in a systematic way

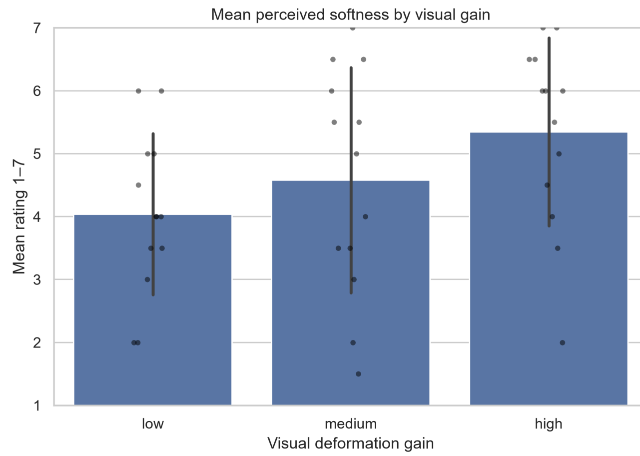

Fig. 2: Perceived deformation clarity by deformation gain level.

The test statistic is reported as χ^2. A higher value indicates a stronger difference between the ranked conditions: the larger this value is, the lower the likelihood that the value is random; this probability is instead defined as the $p-value$. To evaluate on a scale from 0 to 1 how strong the effect of deformation gain is, the effect size Kendall's W was used; values closer to 1 indicating a stronger effect.

TABLE I: Friedman Test - Results

Factor	χ^2	***df***	***p***	***Kendall's W***
Deformation Level	11.91	2	.003	0.4581

The Friedman test showed a significant effect of deformation gain on perceived deformation clarity. This indicates that the three gain levels produced **systematically different perceptual responses**. Kendall's $W = 0.4581$ suggests that the effect was not negligible.

To understand which conditions were different from each other, post-hoc pairwise comparisons were performed. The first post-hoc analysis was performed in Python using Wilcoxon matched-pairs signed-rank tests. This method compares two conditions at a time. Since three pairwise comparisons were performed, the Bonferroni correction was applied, this multiplies each p-value by the number of comparisons made to reduce the risk of false positive results.

Wilcoxon pairwise comparisons confirmed that the **high-gain condition** differed significantly from both **low** and **medium**, while **low** vs. **medium** showed no significant difference. This suggests that the high deformation gain was the condition that produced the clearest perceptual change.

TABLE II: Wilcoxon Comparisons - Results

Comparison	***Statistic***	***p***	***p Bonf.***	***Significant***
low vs medium	11.5	.105	.316	False
medium vs high	6.0	.0078	.023	True
low vs high	0.0	.0019	.0059	True

The same participant-level dataset was also analyzed in JASP as an extended Conover post-hoc comparisons. These second post-hoc showed the same pattern observed in Python: the high gain condition differed from both the low and medium conditions, while the low vs medium comparison was not significant.

TABLE III: Conover Comparisons - Synthesized Results

Comparison	***T-Stat***	***rrb***	***p (Holm)***	***Significant***
low vs. medium	1.376	-0.5818	.182	False
low vs. high	4.402	-1.0000	$<.001$	True
medium vs. high	3.027	-0.8462	.012	True

[a]rrb = rank-biserial correlation.

Note that Table II applies Bonferroni correction while Table III applies Holm correction; despite the different methods, both confirm the same pattern of significance.

To summarize, the post-experiment ratings (Q1–Q7, 1–7 scale) were generally moderate across all dimensions. Participants perceived the ball as more of an interactive object than a mere animation ($M = 5.15$, $SD = 1.46$, Q6), and movement matching was rated moderately well ($M = 4.38$, $SD = 1.26$, Q7).

Subsequently, the threshold that 'breaks' the illusion was analyzed, checking whether the visual deformation felt disconnected from the movement of the hand. The thumb contribution received the lowest mean score ($M = 3.77$, $SD = 2.05$, Q4), consistent with the disconnection data showing the thumb as the most critical moment for the illusion.

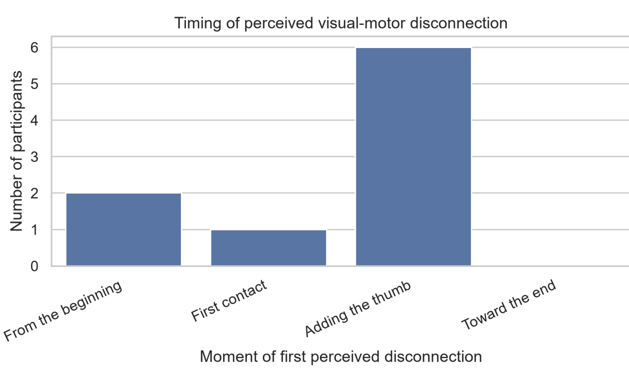

Fig. 3: Disconnection timing perceived from user

This representation is based on the 9 participants out of 13 who reported a disconnection. The most critical moment was the addition of the thumb. A possible explanation is the modeling of the gesture itself or that the thumb made the gesture more similar to a real squeeze, but it also increased the expectation of physical contact. Since the interaction happened in mid-air and the virtual ball could not physically stop or constrain the hand, the thumb could make the absence of a real physical boundary more visible. Further tests would be needed to confirm this interpretation.

VI. Discussion and Future Developments

The results suggest that visual deformation can contribute to pseudo-haptic sensations using only optical hand tracking. The significant effect observed in the trial ratings shows that the perception of softness in XR can be altered by modifying the C/D ratio through visual deformation gain.

Accurately reproducing physical behavior is inherently challenging in mid-air AR, where no real object constrains the hand. For this reason, a soft ball was selected as the virtual object. A soft object allows a certain level of visual deformation and makes the interaction more coherent with the expected behavior of the object. In this way, the experiment supports the idea that visual deformation can modify the sensory-motor expectations of the user.

However, the addition of the thumb represented a critical point for the credibility of the interaction. Adding the thumb made the gesture closer to a real squeeze, but also increased the expectation of physical resistance, exposing the absence of a real boundary. Since the virtual ball could not physically constrain the hand, the thumb sometimes appeared to penetrate the ball or to deform it in an inaccurate way. This **visual-motor mismatch** can weaken or even break the illusion.

Despite the positive results, three main limitations must be addressed for future iterations:

- **Perceptual break during thumb contact:** Descriptive data showed that the perceived disconnection often occurred when the thumb was added. This suggests that thumb contact is one of the most fragile parts of the interaction. When the thumb appears to penetrate the ball or produces an inaccurate deformation, the user can easily detect the discrepancy and the illusion can become weaker.
- **Tremor of optical tracking and self-occlusion:** The optical tracking of the Magic Leap 2 showed some instability during closed-hand gestures. When the thumb overlaps with the index finger or with other parts of the hand, partial self-occlusion can occur. This can generate temporary noise in the tracking data and limit the fluidity of the mesh response.
- **Sample characteristics:** The current sample was useful for a first pilot study, but it was not balanced to perform rigorous statistical comparisons between participant groups. The collected participant information was used only to describe the sample.

For future developments, the experiment could be extended to a complete squeeze movement and to both hands. This would make the interaction closer to the real gesture of compressing a soft ball. Another possible improvement would be to introduce a more advanced modeling, to re-create a more convincing illusion of the ball movement, especially for the thumb, and to reduce the visual penetration of the fingers and thumb into the virtual object.

The research could also be expanded at the psychophysical level. With a more complete interaction and a larger set of deformation gain values, the current rating scale could be replaced or combined with a Two-Alternative Forced Choice (2AFC) paradigm. This would make it possible to estimate the Just Noticeable Difference (JND) and the Point of Subjective Equality (PSE) for compliance perception in optical hand tracking environments. This future analysis would help define more precisely the boundaries within which visual deformation can influence somatic perception before generating a visual-motor disconnection.

Overall, the findings support the idea that visual deformation alone can modulate perceived softness in OST AR, although the illusion remains sensitive to tracking instability and thumb-related interactions.

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